Capacity admission without unnecessary state loss

Conceptual example: 16 usable pages, one ready call A requiring 5 pages, and no earlier pending claim

Ready A outranks optional owner C. Running pages are locked; C is private, idle, and safe to trim.

Current snapshot

running (6) optional C (8) free (2)

Revoke C's 8 pages, then grant A; the checked plan leaves 5 pages free.

Reference plan

running (6) grant A (5) free (5)

Keep C's useful 5-page prefix; release only its final 3 pages before granting A.

Safe partial retention

running (6) grant A (5) retained C (5)

Retention budget U = min(final free − claim credit, minimum commit slack) = min(5 − 0, 5) = 5 pages

Same execution choice

A receives the original grant.

Preserve capacity obligations

Earlier claims and commit limits hold.

Release only safe state

No running, shared, or DMA-held victims.